THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 2:3	Divide both terms by HCF 6
2	**(c)** 12	Multiplier 4, so $x = 3 \times 4$
3	**(b)** 250	One part $= 400 \div 8 = 50$
4	**(a)** 35	5×7
5	**(b)** 36	10% of $360^\circ = 36^\circ$
6	**(c)** 6	Workers × days is constant: $4 \times 12 = 48$
7	**(c)** 3:5	Divide both terms by HCF 5
8	**(b)** 30	Multiplier 6, so $x = 5 \times 6$
9	**(d)** 200	One part $= 360 \div 9 = 40$
10	**(c)** 60	10×6

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cost of 1 pen $=$ ₹15 (1); cost of 15 pens $=$ ₹225 (1).	1+1
12	Men × days $= 10 \times 24 = 240$ (1); days $= 240 \div 8 = 30$ (1).	1+1
13	Total parts $= 7$, one part $=$ ₹120 (1); A gets ₹ 360 and B gets ₹ 480 (1).	1+1
14	Cost of 1 pen $=$ ₹15 (1); cost of 20 pens $=$ ₹300 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Actual distance $= 5 \times 12 = 60$ km (2); map distance $= 96 \div 12 = 8$ cm (1).	2+1
16	Total parts $= 10$, one part $=$ ₹120 (1); A = ₹ 240 (1); B = ₹ 360, C = ₹ 600 (1).	1+1+1
17	Angle $= 30\% \times 360^\circ = 108^\circ$ (2); students $= 60$ (1).	2+1
	OR: Actual distance $= 4 \times 30 = 120$ km (2); map distance $= 150 \div 30 = 5$ cm (1).	2+1
18	Total parts $= 6$, one part $=$ ₹150 (1); A = ₹ 150 (1); B = ₹ 300, C = ₹ 450 (1).	1+1+1
	OR: Angle $= 25\% \times 360^\circ = 90^\circ$ (2); students $= 60$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Angle per student $= 360^\circ \div 30 = 12^\circ$ (1); Cricket: 144°, Football: 108° (2); Tennis: 72°, Hockey: 36° (1); sum $= 360^\circ$ (1).	1+2+1+1
	OR: Total food $= 20 \times 15 = 300$ person-days (2); people now $= 30$ (1); days $= 300 \div 30 = 10$ (2).	2+1+2
20	Angle per student $= 360^\circ \div 45 = 8^\circ$ (1); Maths: 120°, Science: 96° (2); English: 72°, Hindi: 72° (1); sum $= 360^\circ$ (1).	1+2+1+1
	OR: Total food $= 30 \times 12 = 360$ person-days (2); people now $= 36$ (1); days $= 360 \div 36 = 10$ (2).	2+1+2

Mistakes to avoid:

- Treating inverse proportion as direct: more workers means **fewer** days, so use $x_1 y_1 = x_2 y_2$.
- Not converting the map scale to the same units (cm to km) before computing.
- Dividing instead of multiplying (or the other way) when converting map distance to actual distance.
- Computing the pie-chart angle as $\frac{\text{value}}{\text{total}} \times 100$ (percentage) instead of $\times 360^\circ$.
- Using direct proportion for quantities that are not proportional (e.g. age and height).
- Not checking that all sector angles add up to 360°.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 54	15% of $360^\circ = 54^\circ$
2	**(d)** 5	Workers × days is constant: $6 \times 10 = 60$
3	**(a)** 2:3	Divide both terms by HCF 12
4	**(b)** 21	Multiplier 3, so $x = 7 \times 3$
5	**(c)** 330	One part $= 550 \div 5 = 110$
6	**(a)** 100	25×4
7	**(c)** 72	20% of $360^\circ = 72^\circ$
8	**(c)** 6	Workers × days is constant: $8 \times 15 = 120$
9	**(d)** 5:7	Divide both terms by HCF 7
10	**(c)** 30	Multiplier 5, so $x = 6 \times 5$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Men × days $= 15 \times 16 = 240$ (1); days $= 240 \div 12 = 20$ (1).	1+1
12	Total parts $= 13$, one part $= ₹50$ (1); A gets ₹ 300 and B gets ₹ 350 (1).	1+1
13	Cost of 1 pen $= ₹35$ (1); cost of 12 pens $= ₹420$ (1).	1+1
14	Men × days $= 6 \times 20 = 120$ (1); days $= 120 \div 15 = 8$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Actual distance $= 9 \times 8 = 72$ km (2); map distance $= 56 \div 8 = 7$ cm (1).	2+1
16	Total parts $= 15$, one part $= ₹100$ (1); A = ₹ 300 (1); B = ₹ 500, C = ₹ 700 (1).	1+1+1
17	Angle $= 40\% \times 360^\circ = 144^\circ$ (2); students $= 60$ (1).	2+1
	OR: Actual distance $= 6 \times 40 = 240$ km (2); map distance $= 280 \div 40 = 7$ cm (1).	2+1
18	Total parts $= 9$, one part $= ₹200$ (1); A = ₹ 400 (1); B = ₹ 600, C = ₹ 800 (1).	1+1+1
	OR: Angle $= 15\% \times 360^\circ = 54^\circ$ (2); students $= 60$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Angle per student $= 360^\circ \div 50 = 7.2^\circ$ (1); Bus: 144°, Car: 100.8° (2); Cycle: 57.6°, Walk: 57.6° (1); sum $= 360^\circ$ (1).	1+2+1+1
	OR: Total food $= 24 \times 10 = 240$ person-days (2); people now $= 30$ (1); days $= 240 \div 30 = 8$ (2).	2+1+2
20	Angle per student $= 360^\circ \div 30 = 12^\circ$ (1); Red: 120°, Blue: 96° (2); Green: 72°, Yellow: 72° (1); sum $= 360^\circ$ (1).	1+2+1+1
	OR: Total food $= 45 \times 8 = 360$ person-days (2); people now $= 60$ (1); days $= 360 \div 60 = 6$ (2).	2+1+2

Mistakes to avoid:

- Treating inverse proportion as direct: more workers means **fewer** days, so use $x_1 y_1 = x_2 y_2$.
- Not converting the map scale to the same units (cm to km) before computing.
- Dividing instead of multiplying (or the other way) when converting map distance to actual distance.
- Computing the pie-chart angle as $\frac{\text{value}}{\text{total}} \times 100$ (percentage) instead of $\times 360^\circ$.
- Using direct proportion for quantities that are not proportional (e.g. age and height).
- Not checking that all sector angles add up to 360°.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 400	One part $= 700 \div 7 = 100$
2	**(b)** 180	20×9
3	**(d)** 90	25% of $360^\circ = 90^\circ$
4	**(d)** 12	Workers × days is constant: $10 \times 18 = 180$
5	**(c)** 2:5	Divide both terms by HCF 9
6	**(a)** 32	Multiplier 4, so $x = 8 \times 4$
7	**(c)** 800	One part $= 900 \div 9 = 100$
8	**(a)** 150	50×3
9	**(b)** 126	35% of $360^\circ = 126^\circ$
10	**(a)** 15	Workers × days is constant: $12 \times 20 = 240$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total parts $= 8$, one part $= ₹120$ (1); A gets ₹ 600 and B gets ₹ 360 (1).	1+1
12	Cost of 1 pen $= ₹12$ (1); cost of 25 pens $= ₹300$ (1).	1+1
13	Men × days $= 14 \times 18 = 252$ (1); days $= 252 \div 21 = 12$ (1).	1+1
14	Total parts $= 5$, one part $= ₹200$ (1); A gets ₹ 400 and B gets ₹ 600 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Actual distance $= 7 \times 18 = 126$ km (2); map distance $= 126 \div 18 = 7$ cm (1).	2+1
16	Total parts $= 7$, one part $= ₹300$ (1); A = ₹ 300 (1); B = ₹ 1200, C = ₹ 600 (1).	1+1+1
17	Angle $= 35\% \times 360^\circ = 126^\circ$ (2); students $= 126$ (1).	2+1
	OR: Actual distance $= 3 \times 35 = 105$ km (2); map distance $= 245 \div 35 = 7$ cm (1).	2+1
18	Total parts $= 11$, one part $= ₹150$ (1); A = ₹ 450 (1); B = ₹ 600, C = ₹ 600 (1).	1+1+1
	OR: Angle $= 20\% \times 360^\circ = 72^\circ$ (2); students $= 36$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Angle per student $= 360^\circ \div 40 = 9^\circ$ (1); Rice: 162°, Wheat: 108° (2); Maize: 54°, Pulses: 36° (1); sum $= 360^\circ$ (1).	1+2+1+1
	OR: Total food $= 36 \times 10 = 360$ person-days (2); people now $= 60$ (1); days $= 360 \div 60 = 6$ (2).	2+1+2
20	Angle per student $= 360^\circ \div 60 = 6^\circ$ (1); Tea: 144°, Coffee: 108° (2); Milk: 72°, Juice: 36° (1); sum $= 360^\circ$ (1).	1+2+1+1
	OR: Total food $= 15 \times 16 = 240$ person-days (2); people now $= 20$ (1); days $= 240 \div 20 = 12$ (2).	2+1+2

Mistakes to avoid:

- Treating inverse proportion as direct: more workers means **fewer** days, so use $x_1 y_1 = x_2 y_2$.
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